

2.8 Dosage Regimen

Introduction

- Dosage regimen
 - This is a formalized schedule by which drugs are administered,
 - It includes the dosage form, dosing rate, dose, dose interval between doses, route of administration, and duration of therapy
- The success of drug therapy is highly dependent on the choice of the drug and drug product and on the design of the dosage regimen.



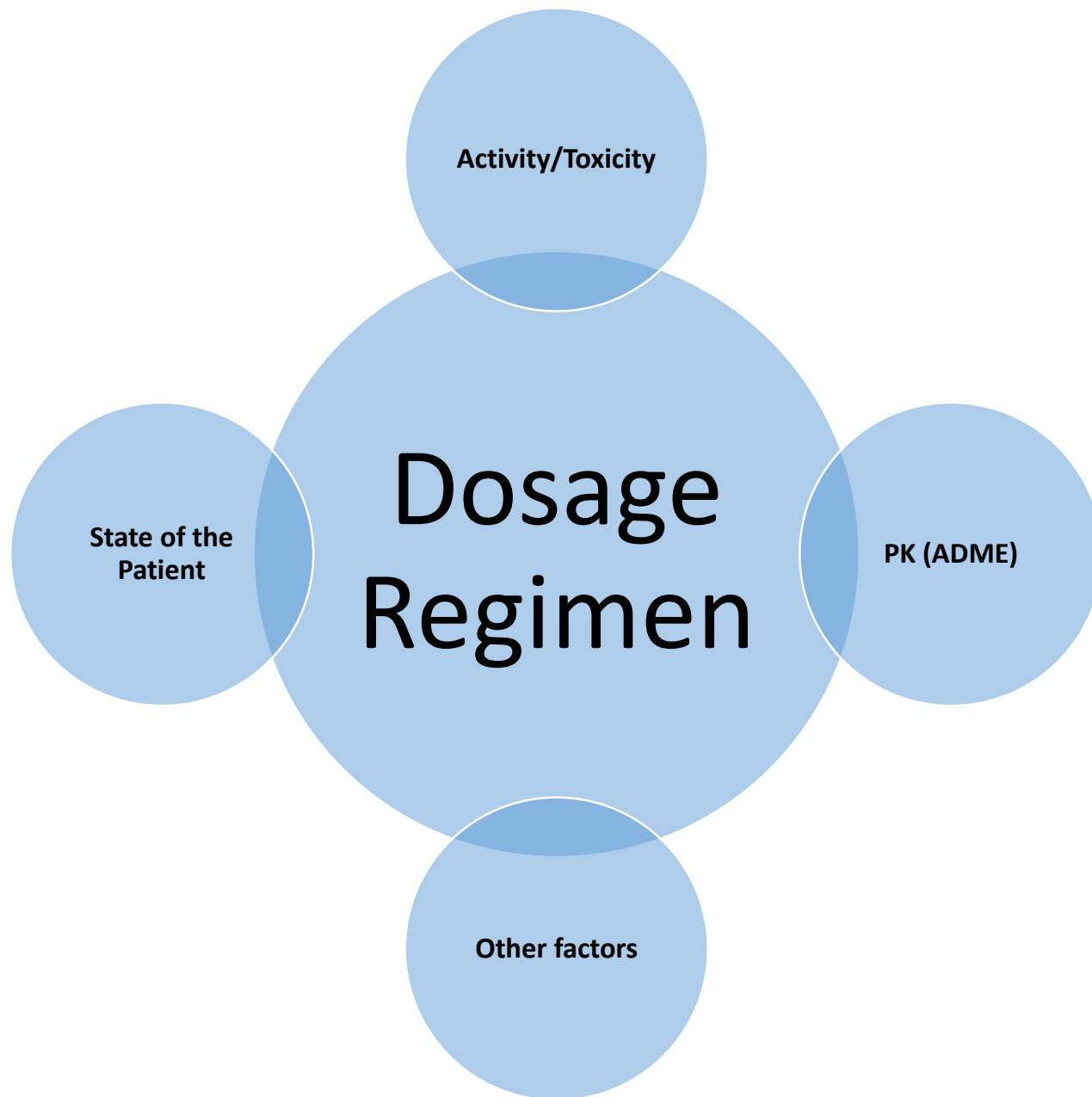
- Therapeutics must be dominated by objective evaluation of an adequate base of factual knowledge.
 - This philosophy has been popularized recently under the terminology of “**evidence-based medicine**”.
- In clinical practice, pharmacokinetic principles are used to characterize the relationships between drug dosage regimens and drug concentration–time profiles.



- When starting a patient on a particular drug therapy, the clinician is dependent on general dosing information.
 - Here, the clinician cannot know exactly how this specific patient will react to the drug, Pharmacokinetically or Pharmacodynamically.
- Optimal therapeutic outcome is a term that has been used to describe “the right drug, to the right patient, at the right time”



- Different factors affect drug response (drug selection).
 - Pharmacodynamic factors
 - Pharmacokinetic factors (Factors affecting ADME processes)
 - Patient related factors
- Dosage regimens are generally derived using average pharmacokinetic parameters to maintain plasma concentrations of the drug within the therapeutic window.
 - The above factors must be taken into consideration for maintaining desired therapeutic concentrations of a drug





- Dosage regimens are designed to produce a therapeutic objective.
 - This objective may be achieved by various modalities of drug administration, extending from a single occasional dose to continuous and constant input.
- In other words, treatment of patients with various disease or conditions with drugs often requires the maintenance of systemic exposure.



- Treatment of patients with various disease or conditions with drugs often requires the maintenance of systemic exposure,
- To do so, drug may be given by:
 - Constant-rate protocol (Constant-rate Regimen)
 - Repeated administration (Multiple-dosage Regimen)
- A therapeutic concentration is appropriately defined in terms of an individual patient's requirement.
 - Usually, this information is unknown on initiating therapy.

Constant-rate Regimen

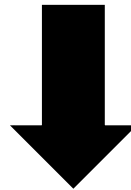
- A single dose may rapidly produce a desired therapeutic concentration,
 - But this mode of administration is unsuitable when maintenance of plasma or tissue concentrations and effect is desired.
- To maintain a constant plasma concentration, drug must be administered at constant rate.
- Could be done with
 - Constant rate intravenous infusion or
 - Constant rate release devices (extra-vascular)



- A wider application of constant-rate therapy become possible with the development and use of constant-rate release devices.
- Revise the calculation for the different PK parameters during:
 - Constant rate infusion
 - IV bolus + Constant rate infusion

Multiple-dosage Regimen

- Although constant-rate regimens possess many desirable features, they are not the most common ones.
- The more common approach to the maintenance of continuous therapy is to give multiple discrete doses.

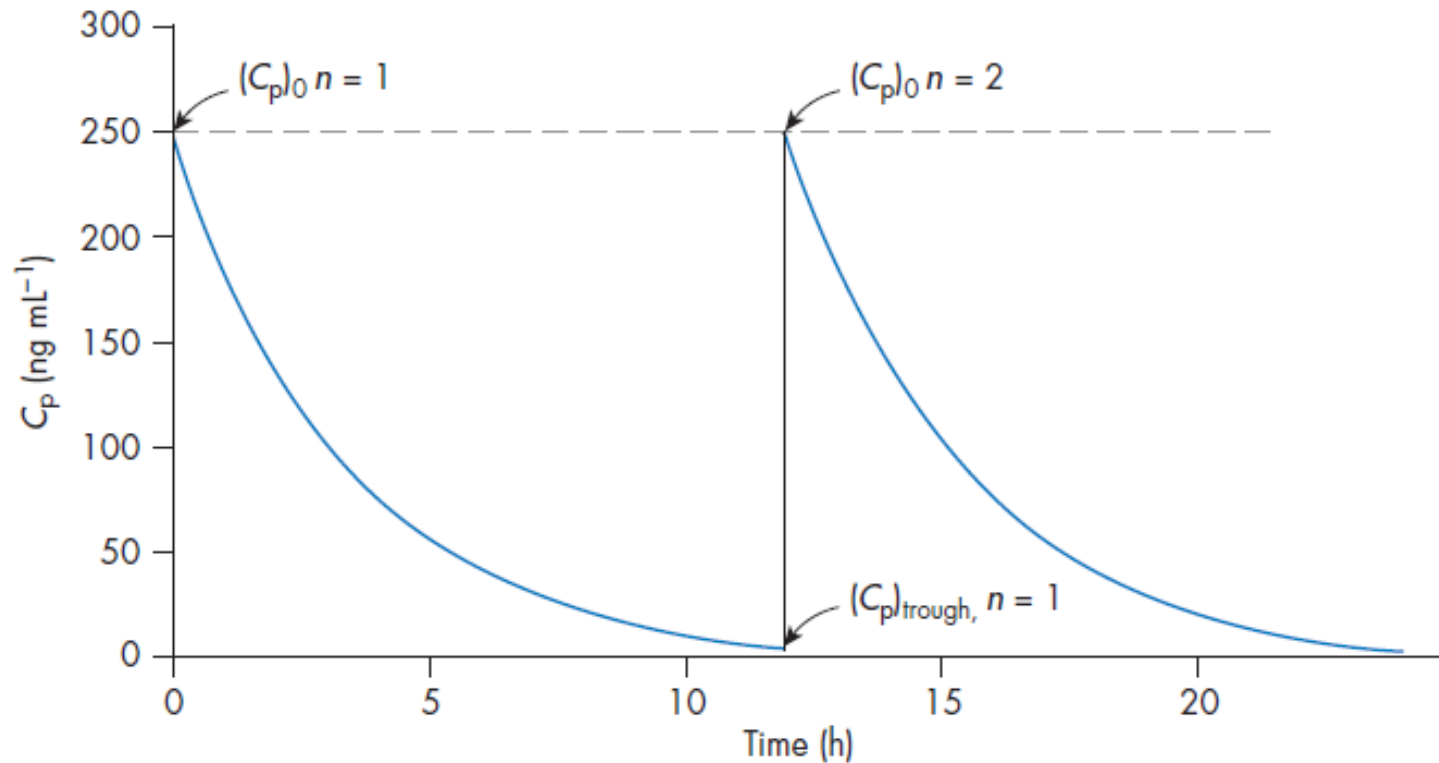


Multiple dosage regimen

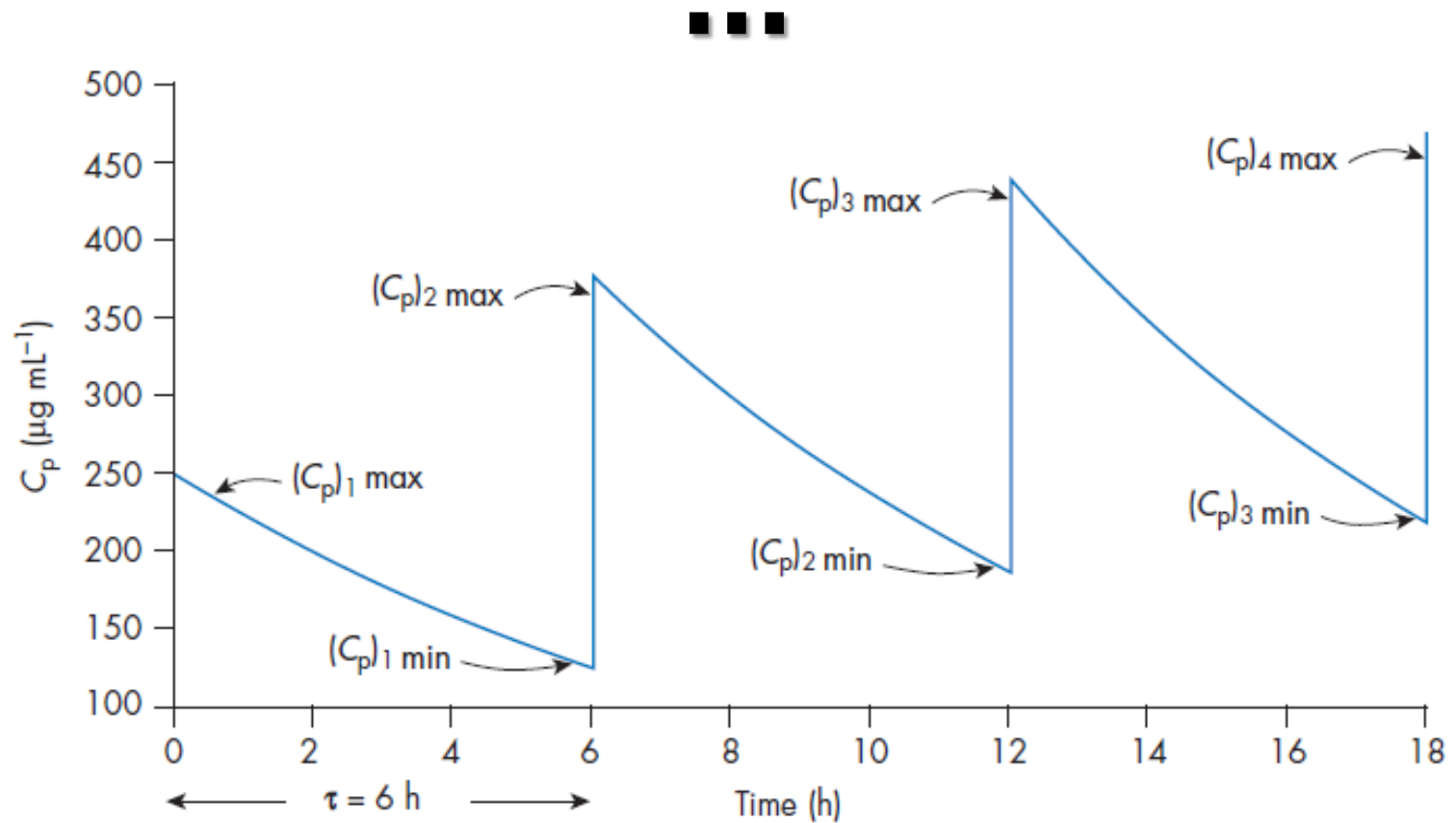


Multiple dosing concepts

- Peak plasma level, following the administration of each succeeding dose, is higher than the peak concentration following the administration of the first dose.
- This phenomenon is the consequence of drug **accumulation** in the body relative to the first dose.
- Following the administration of a number of doses, a steady state condition will be achieved.



Next dose is administered when there is essentially no drug accumulation in the body; in other words, there is no drug left in the body from previous doses.



Next doses are administered at finite time intervals.



- In obtaining the different parameters for multiple dosing, the following theoretical **assumptions** are made:
 - Linear pharmacokinetics applies; that is, the rate process obeys passive diffusion and first order elimination kinetics.
 - Tissues can take up an infinite amount of drug, if necessary.
 - The apparent volume of distribution (V), elimination half life ($t_{1/2}$) and the elimination rate constant (K) are independent of the number of administered doses.



- The time interval between dosing or successive doses is constant.
- The administered dose is equal at each successive time interval.



- Useful pharmacokinetic parameters in multiple dosing include:
 - The Dost ratio (r)
 - The concentration of drug in the body at any time t during the dosing interval.
 - The steady-state plasma concentrations.
 - The maximum and minimum plasma concentrations at steady state.



- The “average” steady-state plasma concentration,
- Drug accumulation (R),
- Fluctuation (Φ),
- Number of doses (n) required to attain a given fraction of steady state (f_{ss})
- Loading (D_L) and maintenance dose (D_M)

Multiple-dosage Regimen: IV Administration

A single IV bolus dose

- Over time after an IV bolus dose, the amount and concentration of the drug in the body is given by:

$$X = X_0 e^{-Kt}$$

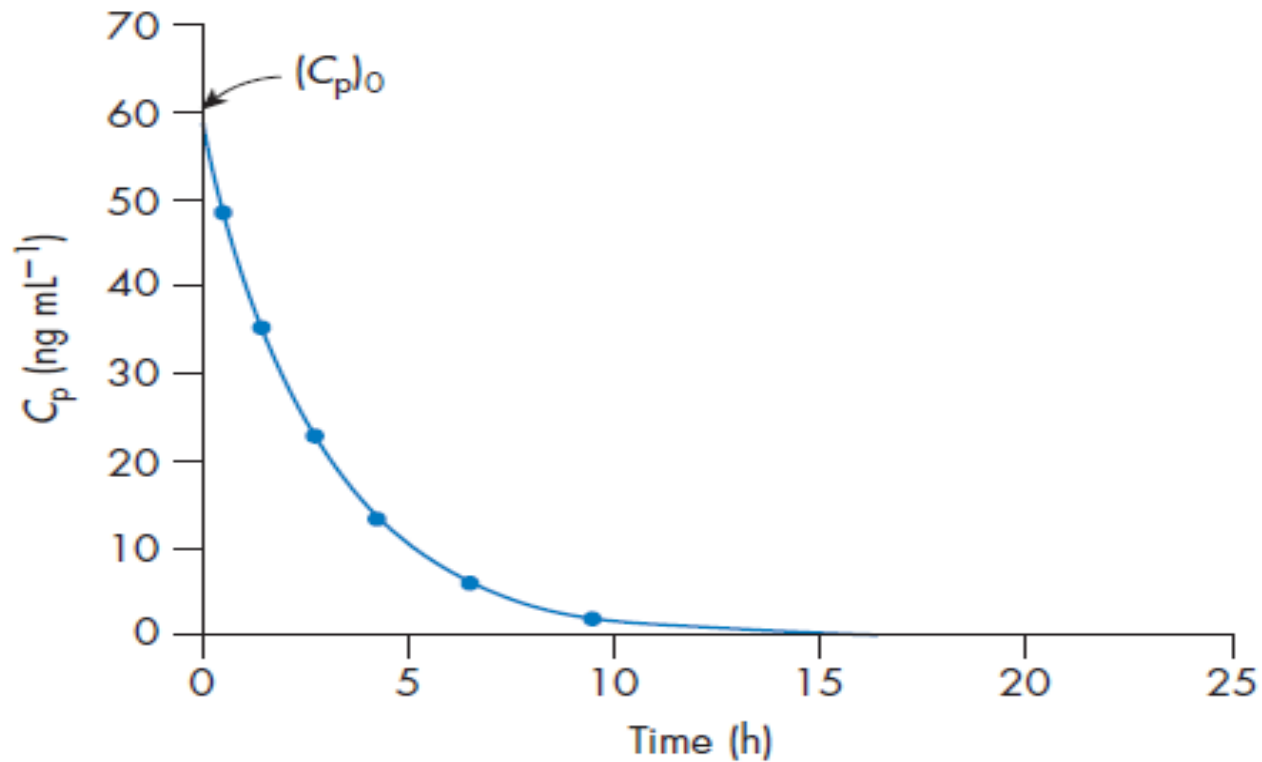
And,

$$C_p = (C_p)_0 e^{-Kt}$$



- From such administration;
 - At $t = 0$,
 - $X = X_0$
 - $C_p = (C_p)_0$ (i.e. maximum or peak plasma concentration, or intercept of the concentration versus time plot) and,
 - At $t = \infty$
 - $X = 0$ and $C_p = 0$.

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A typical plasma concentration versus time profile for a single IV bolus dose of a drug that follows first-order elimination and has one compartment distribution.



Multiple IV bolus dose

The Dost ratio (r)

- The Dost ratio permits the determination of the X and/or the C_p of a drug in the body at any time t following the administration of the n^{th} dose.
- In other words, this ratio will transform a single dose equation into a multiple-dosing equation.

$$r = \frac{1 - e^{-nK\tau}}{1 - e^{-K\tau}}$$

Where, n is number of administered doses; K is the first-order elimination rate constant; and τ is the dosing interval



Plasma concentration at any time t: $(C_{pn})_t$

- To determine the plasma concentration of a drug in the body at any time, t, following administration of the n^{th} dose:
 - Inserting the Dost ratio between $(C_p)_0$ and e^{-Kt} in the equation $C_p = (C_p)_0 e^{-Kt}$

$$(C_{pn})_t = (C_p)_0 \frac{(1 - e^{-nK\tau})}{(1 - e^{-K\tau})} e^{-Kt}$$



- From the above equation,

- When $n = 1$, (i.e. administration of the first dose)

$$(C_{pn})_t = (C_p)_0 \frac{(1 - e^{-nK\tau})}{(1 - e^{-K\tau})} e^{-Kt} \quad \rightarrow \quad C_p = (C_p)_0 e^{-Kt}$$

- When $n = \infty$ (i.e. administration of many doses; generally more than eight or nine), $e^{-nK\tau} = 0$

- **Note:** t in multiple dose PK is the time since the latest (n^{th}) dose was given (i.e. $t \geq 0$ and $t < \tau$) (a value between dosing interval).



- At $t = 0$, $e^{-Kt} = 1$,
- The plasma concentration at this time is the maximum plasma concentration $(C_p)_{max}$.
- Therefore,

$$(C_{pn})_{max} = \frac{(C_p)_o (1 - e^{-nK\tau})}{(1 - e^{-K\tau})}$$

- This Eq. represents the maximum plasma concentration of a drug in the body following the administration of the n^{th} dose.
- Referred as “**peak**” plasma concentration.



- At $t = \tau$,
- The body will display the minimum plasma concentration, $(C_p)_{min}$, of a drug.

$$(C_{pn})_{min} = \frac{(C_p)_o (1 - e^{-nK\tau})}{(1 - e^{-K\tau})} e^{-K\tau}$$

- This Eq. represents the minimum plasma concentration of a drug in the body following administration of the n^{th} dose.
- Referred as “**trough**” plasma concentration.



Steady-state plasma concentration

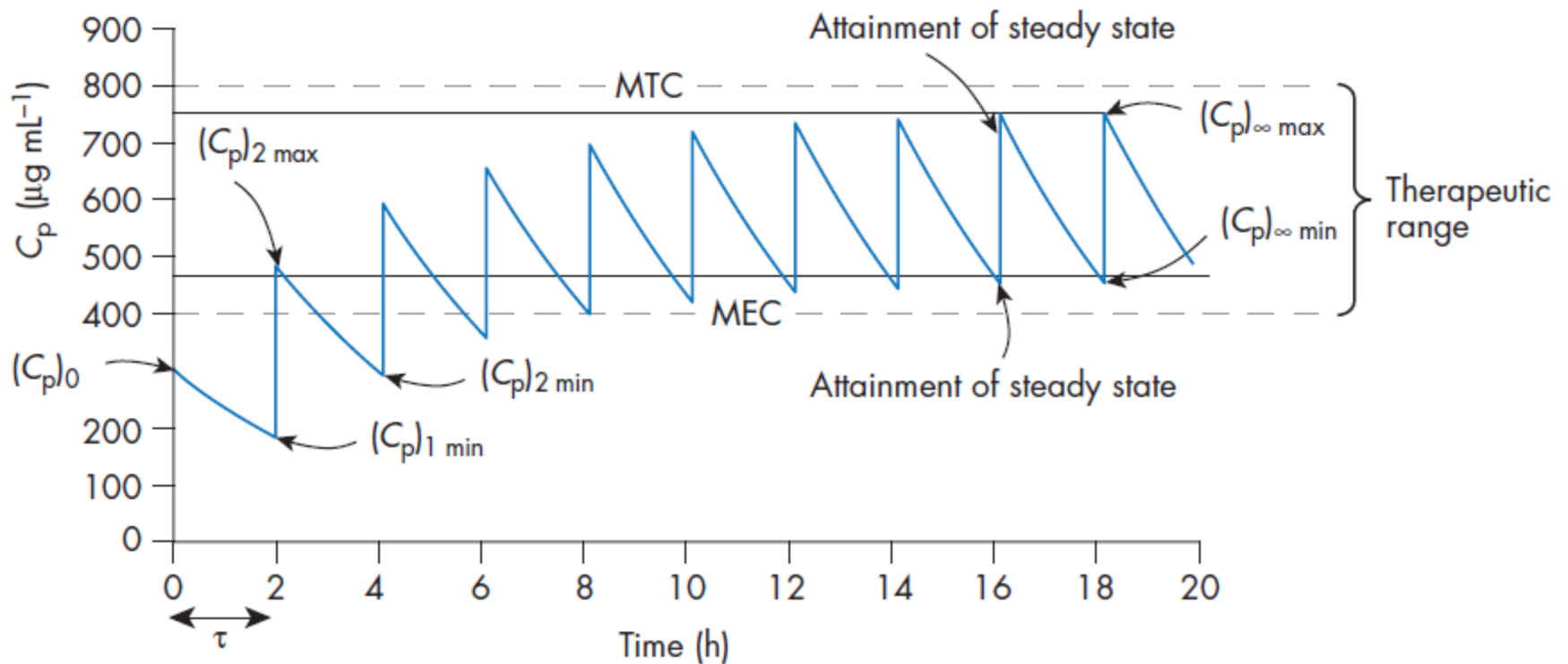
- The steady-state condition is attained following the administration of many doses (i.e. $n = \infty$).
- When $n \rightarrow \infty$, $e^{-nK\tau} \rightarrow 0$, then

$$(C_{pn})_t = \frac{(C_p)_0 (1 - e^{-nK\tau})}{(1 - e^{-K\tau})} e^{-Kt} \rightarrow (C_{p\infty})_t = \frac{(C_p)_0}{(1 - e^{-K\tau})} e^{-Kt}$$

- This Eq. represents the plasma concentration of drug in the body at any time t following the attainment of steady state.



The $(C_{pn})_{max}$ and $(C_{pn})_{min}$ at steady state





- For drugs administered IV,
 - The maximum steady-state plasma concentrations will occur only at $t = 0$ (immediately after administration of the latest bolus dose).

$$(C_{p\infty})_{max} \text{ or } (C_{pss})_{max} = \frac{(C_p)_0}{(1 - e^{-K\tau})}$$

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- And, the minimum steady-state plasma concentrations will occur only at $t = \tau$ (immediately after administration of the latest bolus dose).

$$(C_{p\infty})_{min} \text{ or } (C_{pss})_{min} = \frac{(C_p)_0}{(1 - e^{-K\tau})} e^{-K\tau}$$

Or

$$(C_{pss})_{min} = (C_{pss})_{max} e^{-K\tau}$$

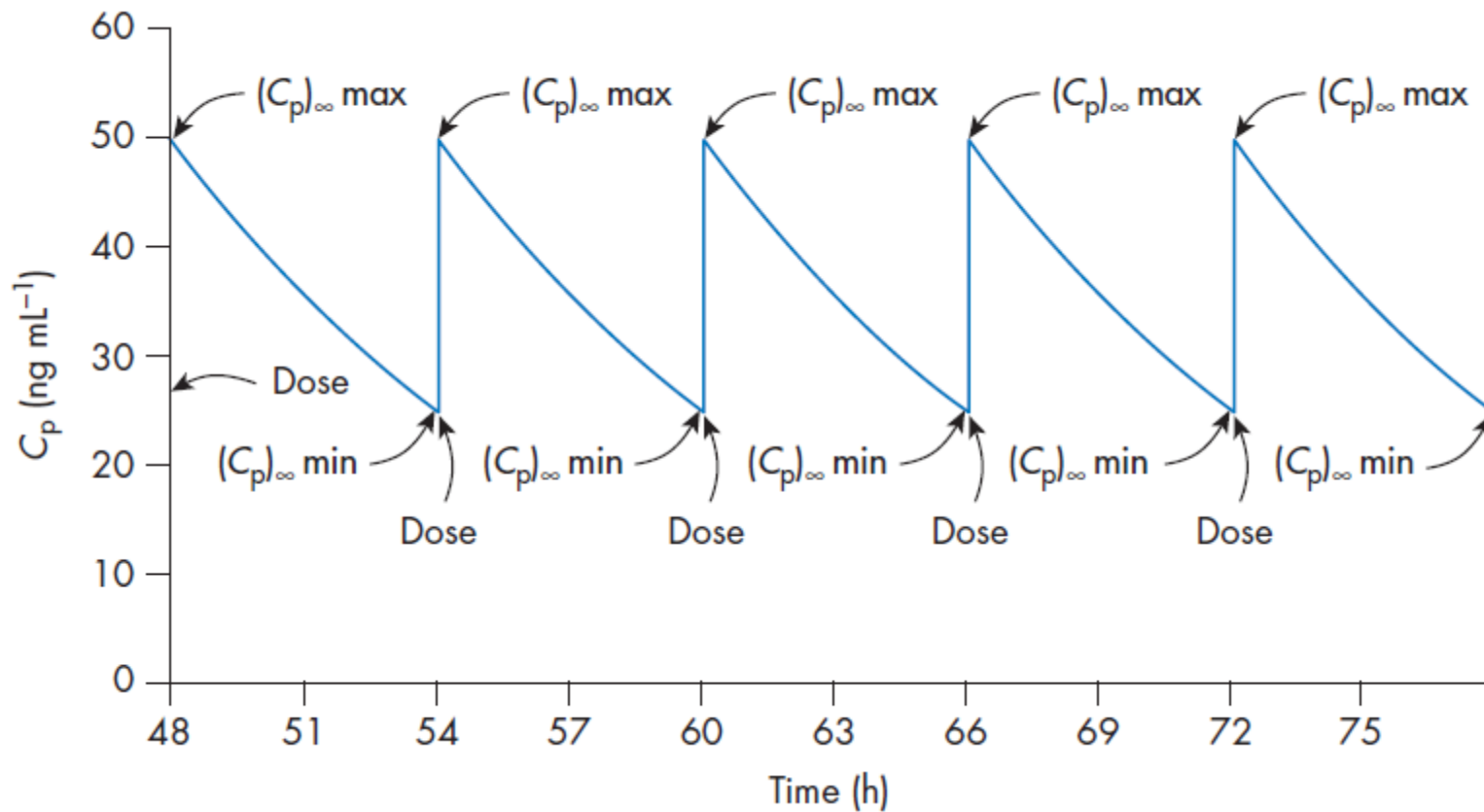


Fig. C_p vs t profile following the attainment of steady-state condition for a drug administered by multiple intravenous bolus injections.



The “Average” plasma concentration at steady state

- A very useful parameter in multiple dosing.
- This plasma concentration value will fall between the maximum and minimum steady-state plasma concentration values.
 - It is the one desired therapeutically for patients on a regular dosage regimen.

$$(\bar{C}_p)_{ss} = \frac{\int_0^{\tau} C_{p\infty} dt}{\tau}$$

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- The term $\int_0^\tau C_{p_\infty} dt$ is the AUC at steady state during dosing interval.

$$AUC = \int_0^\tau C_{p_\infty} dt = \frac{X_o}{VK}$$

- Therefore,

$$(\bar{C}_p)_{ss} = \frac{X_o}{VK\tau}$$



- Designing or establishing the dosage regimen for a drug:
 - Know the therapeutic range and/or the effective concentration range for the drug;
 - Select the desired or targeted “average” steady-state plasma concentration (commonly the mean of the therapeutic range);
 - Select the dosing interval (it is a safe and good practice to start with a dosing interval equal to the drug’s elimination half life).
 - Calculate the dose (X_0) needed to attain the desired “average” steady-state concentration.



- Then, calculate the “average” steady-state concentration, peak steady-state concentration and trough steady-state concentration;
- Make sure that the calculated peak steady-state concentration is below the minimum toxic concentration and calculated trough steady-state concentration is above the minimum effective concentration;
- If necessary, make small adjustments (fine tuning) in the dose and dosing interval.



Drug accumulation (R)

- The administration of a drug in a multiple dose regimen will always result in the accumulation of drug in the body.
- The extent of accumulation (R) of a drug may be quantified in several ways.
- During any dosing interval, the “average” plasma concentration of a drug may be defined as:

$$(\bar{C}_p)_n = \frac{\int_0^{\tau} C_{p_n} dt}{\tau}$$



- Substituting $\frac{X_o}{V}$ for $(C_p)_o$ in the equation, $(C_{pn})_t = \frac{(C_p)_o(1-e^{-nK\tau})}{(1-e^{-K\tau})} e^{-Kt}$
- And integrating from $t = 0$ to $t = \tau$, yields:

$$\int_0^{\tau} C_{pn} dt = \frac{X_o}{VK} [1 - e^{-nK\tau}]$$

- Dividing both sides by τ :

$$\frac{\int_0^{\tau} C_{pn} dt}{\tau} = \frac{X_o}{VK\tau} [1 - e^{-nK\tau}]$$

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- Since $(\bar{C}_p)_n = \frac{\int_0^\tau C_{pn} dt}{\tau}$,

$$(\bar{C}_p)_n = \frac{X_o}{VK\tau} [1 - e^{-nK\tau}]$$

- And, $(\bar{C}_p)_{ss} = \frac{X_o}{VK\tau}$

- Therefore,

$$(\bar{C}_p)_n = (\bar{C}_p)_{ss} [1 - e^{-nK\tau}]$$

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- Rearranging the above equation:

$$\frac{(\bar{C}_p)_n}{(\bar{C}_p)_{ss}} = [1 - e^{-nK\tau}]$$

- When $n = 1$

$$\frac{(\bar{C}_p)_1}{(\bar{C}_p)_{ss}} = [1 - e^{-K\tau}]$$



- The inverse ratio, $\frac{(\bar{C}_p)_{ss}}{(\bar{C}_p)_1}$, may be defined as an accumulation factor (R);

$$R = \frac{(\bar{C}_p)_{ss}}{(\bar{C}_p)_1} = \frac{1}{[1 - e^{-K\tau}]}$$

- 'R' indicates the extent to which a drug would accumulate in the body.

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- Use of other ratios to calculate drug accumulation:

$$R = \frac{(C_{p\infty})_{min}}{(C_{p1})_{min}} = \frac{\frac{(C_p)_0}{(1 - e^{-K\tau})} e^{-K\tau}}{(C_p)_0 e^{-K\tau}} = \frac{1}{1 - e^{-K\tau}}$$

- Or

$$R = \frac{(C_{p\infty})_{max}}{(C_{p1})_{max}} = \frac{\frac{(C_p)_0}{(1 - e^{-K\tau})}}{(C_p)_0} = \frac{1}{1 - e^{-K\tau}}$$



Determination of fluctuation (Φ):

- Fluctuation is simply a measure of the magnitude of variation (differences) in the peak and trough plasma concentrations at steady state.
 - It is simply a measure of the ratio of the steady-state maximum plasma concentration to the steady-state minimum, or
 - The ratio of the maximum steady-state concentration to the “average” plasma concentration at steady state for the chosen dosage regimen.

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- By comparison of maximum concentration with minimum concentration at steady state:

$$\Phi = \frac{(C_{pss})_{max}}{(C_{pss})_{min}} = \frac{\frac{(C_p)_o}{(1 - e^{-K\tau})}}{\frac{(C_p)_o}{(1 - e^{-K\tau})} e^{-K\tau}} = \frac{1}{e^{-K\tau}}$$

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- If the dosing interval (τ) is expressed in terms of number (N) of elimination half lives ($t_{1/2}$), $\tau = N t_{1/2}$ and $K = \frac{0.693}{t_{1/2}}$
- Then,

$$\Phi = \frac{1}{e^{-(0.693)(N)}}$$

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Number of doses required to reach steady-state condition

- The plasma concentration at time t after the n^{th} dose is:

$$(C_{pn})_t = \frac{(C_p)_0 (1 - e^{-nK\tau})}{(1 - e^{-K\tau})} e^{Kt}$$

- And the steady-state plasma concentration at time t is:

$$(C_{p\infty})_t = \frac{(C_p)_0}{(1 - e^{-K\tau})} e^{-Kt}$$

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$$\frac{(C_{pn})_t}{(C_{p\infty})_t} = \frac{\frac{(C_p)_o (1 - e^{-nK\tau})}{(1 - e^{-K\tau})} e^{Kt}}{\frac{(C_p)_o}{(1 - e^{-K\tau})} e^{-Kt}} = f_{ss} = 1 - e^{-nK\tau}$$

- For $\tau = N t_{1/2}$ and $K = \frac{0.693}{t_{1/2}}$

$$f_{ss} = 1 - e^{-0.693nN}$$

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- Taking the natural logarithm and rearrangement of the above equation yields:

$$n = - \frac{\ln(1 - f_{ss})}{0.693N}$$

where n is the number of doses required to reach a given fraction of the steady-state (f_{ss}) condition and N is the number of elimination half lives in the dosing interval.



Loading and Maintenance doses

- It may take a long time and the administration of many doses before the desired “average” steady-state drug concentration is attained.
- Therefore, an IV bolus loading dose (D_L) may be administered to obtain an instant steady-state condition.



- The calculated D_L should be such that that, at time t after its administration, the plasma concentration of drug is the desired minimum plasma concentration at steady state, that is:

$$(C_{pss})_{min} = \frac{D_L}{V} e^{-K\tau}$$

- And we know that,

$$(C_{pss})_{min} = \frac{X_o}{V} \frac{e^{-K\tau}}{(1 - e^{-K\tau})}$$

where X_o is the IV bolus maintenance dose (D_M).

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- Then,

$$\frac{D_L}{V} e^{-K\tau} = \frac{D_M}{V} \frac{e^{-K\tau}}{(1 - e^{-K\tau})}$$

- Upon simplification and rearrangement, we get:

$$\frac{D_L}{D_M} = \frac{V e^{-K\tau}}{V(1 - e^{-K\tau}) e^{-K\tau}} = \frac{1}{1 - e^{-K\tau}}$$

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- For $\tau = N t_{1/2}$ and $K = \frac{0.693}{t_{1/2}}$

$$\frac{D_L}{D_M} = \frac{1}{1 - e^{-(0.693)(N)}}$$

Multiple-dosage Regimen: EV Administration

- Extravascular delivery is not instantaneous (drug not immediately delivered into the systemic circulation).
- Therefore, the pharmacokinetic equations require a term reflecting an **absorption** process.



Single extravascular dose

- Over time after an extravascular dose, the amount and concentration of the drug in the body is given by:

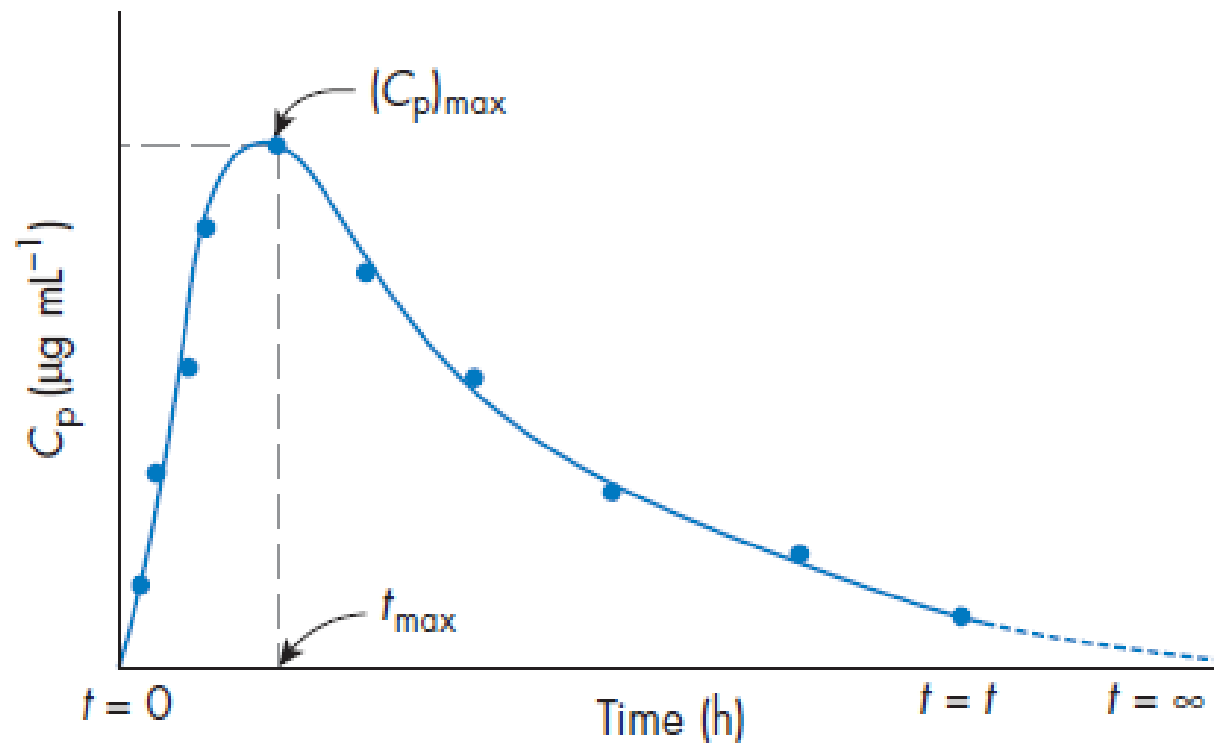
$$(X)_t = \frac{K_a F X_o}{(K_a - K)} (e^{-Kt} - e^{-K_a t})$$

- And,

$$(C_p)_t = \frac{K_a F X_o}{V(K_a - K)} (e^{-Kt} - e^{-K_a t})$$

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- From such administration,
 - At $t = 0$ and $t = \infty$, $X = 0$ and $C_p = 0$
 - at $t = t_{max}$ (i.e. peak time), $X = X_{max}$ and $C_p = (C_p)_{max}$



A typical plasma concentration (C_p) versus time profile for a drug administered as a single dose of drug by an extravascular route.



Multiple Extravascular Dosing

- The plasma concentration versus time curve following multiple dosing of a drug which is absorbed by an apparent first order process can be described as follows;
- For a single dose EV administration:

$$(C_p)_t = \frac{K_a F X_0}{V(K_a - K)} (e^{-Kt} - e^{-K_a t})$$

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- And multiplying each exponential term by “r” yields:

$$(C_{pn})_t = \frac{K_a F X_o}{V(K_a - K)} \left(\left[\frac{1 - e^{-nK\tau}}{1 - e^{-K\tau}} \right] e^{-Kt} - \left[\frac{1 - e^{-nK_a\tau}}{1 - e^{-K_a\tau}} \right] e^{-K_a t} \right)$$

where $(C_{pn})_t$ is the plasma concentration at time t following the administration of the nth dose

- This Eq. may be employed to predict the plasma concentration of drug at any time during the dosing interval following the administration of nth dose.



The peak time in multiple dosing to steady state ($t'_{\max}$)

- $t'_{\max}$ to steady state ($n = \infty$) can be obtained by differentiating the following equation with respect to time and setting the resultant equal to 0.

$$(C_{p\infty})_t = \frac{K_a F X_0}{V(K_a - K)} \left(\left[\frac{1}{1 - e^{-K\tau}} \right] e^{-Kt} - \left[\frac{1}{1 - e^{-K_a\tau}} \right] e^{-K_a t} \right)$$



- Differentiating the above Eq. with respect to t yields:

$$\frac{d(C_{p\infty})_t}{dt} = \frac{K_a F X_0}{V(K_a - K)} \left(\left[\frac{K_a e^{-K_a t}}{1 - e^{-K_a \tau}} \right] - \left[\frac{K e^{-K t}}{1 - e^{-K \tau}} \right] \right)$$

- To find $t = t'_{max}$, we set the above expression to equal zero, and simplification yields:

$$\frac{K_a e^{-K_a t'_{max}}}{1 - e^{-K_a \tau}} = \frac{K e^{-K t'_{max}}}{1 - e^{-K \tau}}$$

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- Further simplification yields:

$$t'_{max} = \frac{2.303}{K_a - K} \log \left[\frac{K_a}{K} \left(\frac{1 - e^{-K\tau}}{1 - e^{-K_a\tau}} \right) \right]$$

- The peak time following the administration of a single extravascular dose is obtained by:

$$t_{max} = \frac{2.303}{K_a - K} \log \left[\frac{K_a}{K} \right]$$



Maximum plasma concentration at steady state

- Steady state plasma concentration for EV multiple dose administration is obtained by:

$$(C_{p\infty})_t = \frac{K_a F X_0}{V(K_a - K)} \left(\left[\frac{1}{1 - e^{-K\tau}} \right] e^{-Kt} - \left[\frac{1}{1 - e^{-K_a\tau}} \right] e^{-K_a t} \right)$$

- And the maximum plasma concentration at steady state is obtained when $t = t'_{max}$

$$(C_{p\infty})_{max} = \frac{K_a F X_0}{V(K_a - K)} \left(\left[\frac{1}{1 - e^{-K\tau}} \right] e^{-Kt'_{max}} - \left[\frac{1}{1 - e^{-K_a\tau}} \right] e^{-K_a t'_{max}} \right)$$

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- From

$$\frac{K_a e^{-K_a t'_{max}}}{1 - e^{-K_a \tau}} = \frac{K e^{-K t'_{max}}}{1 - e^{-K \tau}}$$

- Rearranging for $e^{-K_a t'_{max}}$

$$e^{-K_a t'_{max}} = \left(\frac{1 - e^{-K_a \tau}}{1 - e^{-K \tau}} \right) \frac{K}{K_a} e^{-K t'_{max}}$$

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- Substitution for $e^{-K_a t'_{max}}$ in the Eq. for $(C_{p\infty})_{max}$ and rearranging:

$$(C_{p\infty})_{max} = \frac{FX_0}{V} \left(\left[\frac{1}{1 - e^{-K\tau}} \right] e^{-K t'_{max}} \right)$$



Minimum plasma concentration at steady state

- Steady state plasma concentration for EV multiple dose administration is obtained by:

$$(C_{p\infty})_t = \frac{K_a F X_0}{V(K_a - K)} \left(\left[\frac{1}{1 - e^{-K\tau}} \right] e^{-Kt} - \left[\frac{1}{1 - e^{-K_a\tau}} \right] e^{-K_a t} \right)$$

- And the minimum plasma concentration at steady state is obtained when $t = \tau$

$$(C_{p\infty})_{min} = \frac{K_a F X_0}{V(K_a - K)} \left(\left[\frac{1}{1 - e^{-K\tau}} \right] e^{-K\tau} - \left[\frac{1}{1 - e^{-K_a\tau}} \right] e^{-K_a\tau} \right)$$

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- In the post-absorptive phase, the term $e^{-K_a\tau}$ approaches zero.
- Therefore;

$$(C_{p\infty})_{min} = \frac{K_a F X_o}{V(K_a - K)} \left(\frac{1}{1 - e^{-K\tau}} e^{-K\tau} \right)$$



“Average” plasma concentration at steady state

- The “average” plasma concentration of a drug at steady state for EV administered dose can be calculated from:

$$(\bar{C}_p)_{ss} = \frac{FX_0}{VK\tau}$$

- The “average” plasma concentration can also be obtained from the area under the plasma concentration time curve during a dosing interval, at steady state.



Drug accumulation

- The accumulation factor (R) following the administration of a drug by an EV route can be calculated by;

$$R = \frac{(C_{p\infty})_{min}}{(C_{p1})_{min}}$$

- Therefore, substituting for $(C_{p\infty})_{min}$ and $(C_{p1})_{min}$

$$R = \frac{1}{1 - e^{-K\tau}}$$

- From

$$(C_{pn})_t = \frac{K_a F X_o}{V(K_a - K)} \left(\left[\frac{1 - e^{-nK\tau}}{1 - e^{-K\tau}} \right] e^{-Kt} - \left[\frac{1 - e^{-nK_a\tau}}{1 - e^{-K_a\tau}} \right] e^{-K_a t} \right)$$

- We can obtain $(C_{p1})_{min}$ at $n=1$ and $t=\tau$:

$$(C_{p1})_{min} = \frac{K_a F X_o}{V(K_a - K)} (e^{-K\tau} - e^{-K_a\tau})$$

- And $(C_{p\infty})_{min}$ at $n = \infty$ and $t=\tau$

$$(C_{p\infty})_{min} = \frac{K_a F X_o}{V(K_a - K)} \left(\left[\frac{1}{1 - e^{-K\tau}} \right] e^{-K\tau} - \left[\frac{1}{1 - e^{-K_a\tau}} \right] e^{-K_a\tau} \right)$$



- And, in the post-absorptive phase, the term $e^{-K_a t}$ approaches zero,
- Therefore,

$$(C_{p1})_{min} = \frac{K_a F X_0}{V(K_a - K)} e^{-K\tau}$$

- And

$$(C_{p\infty})_{min} = \frac{K_a F X_0}{V(K_a - K)} \left(\left[\frac{1}{1 - e^{-K\tau}} \right] e^{-K\tau} \right)$$



Fluctuation factor (Φ)

- To calculate fluctuation at steady state:

$$\Phi = \frac{(C_{p_{ss}})_{max}}{(C_{p_{ss}})_{min}}$$

- (check for fluctuation with multiple intravenous bolus administration)



Number of doses to reach a fraction of steady state

- The fraction of the steady state following EV administration can be estimated as:

$$f_{ss} = \frac{\bar{C}_{p_n}}{\bar{C}_{p_{ss}}}$$

- Substituting for $\bar{C}_{p_n}$ and $\bar{C}_{p_{ss}}$

$$f_{ss} = 1 + \frac{K e^{-nK_a t}}{K_a - K} - \frac{K_a e^{-nK\tau}}{K_a - K}$$

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- If $K_a \gg K$,

$$f_{ss} = 1 - e^{-nK\tau}$$

- Rearranging and solving for n ,

$$n = -\frac{\ln(1 - f_{ss})}{0.693N}$$



Loading and Maintenance Dose

- An initial loading dose may be desirable since a long period of time is required to reach steady state for drugs with long half lives.
- The goal is that $(C_{p1})_{min}$ from the loading dose will exactly equal $(C_{p\infty})_{min}$ from the multiple-dosing regimen.



- Hence, equating the two equations:

$$\frac{K_a F(X_O)_{D_L}}{V(K_a - K)} (e^{-K\tau} - e^{-K_a\tau}) = \frac{K_a F(X_O)_{D_M}}{V(K_a - K)} \left[\frac{e^{-K\tau}}{1 - e^{-K\tau}} - \frac{e^{-K_a\tau}}{1 - e^{-K_a\tau}} \right]$$

- Upon rearranging and simplification:

$$(X_O)_{D_L} = (X_O)_{D_M} \left(\frac{1}{(1 - e^{-K_a\tau})(1 - e^{-K\tau})} \right)$$